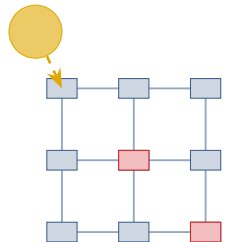


System Model — LEO Satellite Edge Computing with Battery-Health-Aware Offloading

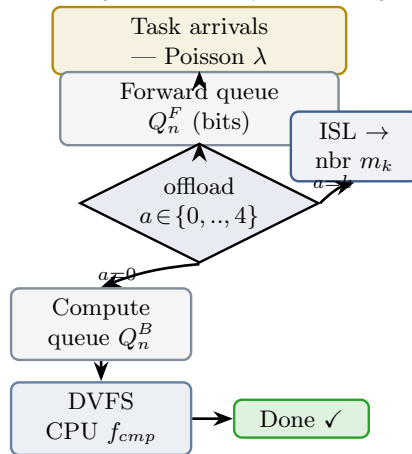
(b) On-board task pipeline (one satellite)

(a) LEO constellation & ISL



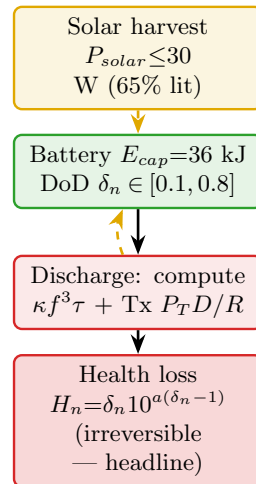
$N=192$ (16×12 Walker);
 3×3 shown ($\dots \times N$)
 ISL: rate $R_{n,m}$, delay
 $\tau_{n,m}$ ($B=100\text{--}300$ Mb/s)
 red = high-load $\lambda_{hi}=4.0$
 blue = low-load $\lambda_{lo}=0.1$

task $K_i = \{D_i \text{ bits}, X_i \text{ cyc/bit}, \text{deadline}\}$



DVFS (Li TSC'24):
 $f_{cmp} = \min(F_{max}, \max(\sqrt{Q^B}/3V_{DVFS}\kappa, f_{floor}));$
 $F_{max}=2 \text{ GHz}, E_{cmp}=\kappa f^3 \tau$

(c) Battery energy cycle



→ task flow

-> energy flow

...> state broadcast (4 neighbours)